



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SANTHOSH AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

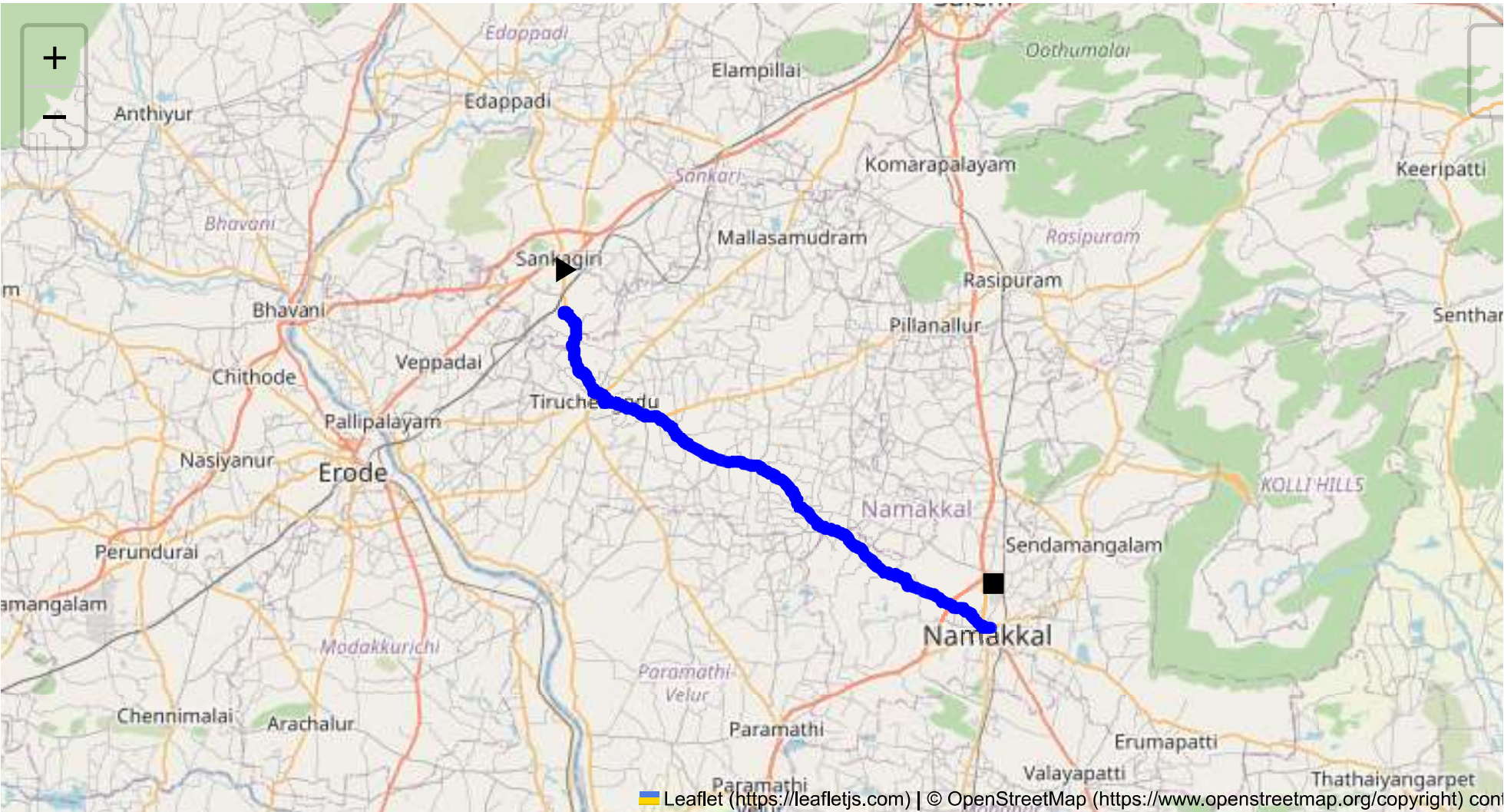
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 44.14 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (11.4381, 77.8734)
End: (11.224396, 78.170158)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to R.P. Pudur traverses approximately 44.14 kilometers mainly along regional highways and roads. This journey takes you through scenic rural areas, with sections that may require navigating around smaller towns or villages. Key roads involved include State Highways and potentially some local access routes.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Conditions:** Tamil Nadu typically experiences a tropical climate. The summer months (March to June) can be extremely hot, while monsoon season (June to September) brings heavy rainfall.
- **Weather Hazards:** During monsoon season, expect potential flooding and slippery roads. Reduced visibility may also occur due to rain or fog.

3. Traffic Patterns

- **Peak Hours:** Morning hours (8 AM to 10 AM) and evening hours (5 PM to 7 PM) could see increased traffic due to commuter travel.
- **Congestion-Prone Areas:** Traffic could be heavier near town centers and intersections, especially where local traffic intersects with the main highway routes.

4. Assessment of Road Quality and Infrastructure

- **General Condition:** The roads, being highways and state roads, are generally maintained well, but conditions can vary with weather impacts such as potholed areas post-monsoon.
- **Infrastructure Quality:** Some areas might lack proper signage or lane demarcations, and lighting might be insufficient in rural stretches, posing visibility challenges at night.

5. Suggestions for Alternative Routes for Emergencies

In case of road closures or incidents:

- **Via SH79 and NH544:** This alternative can offer less traffic congestion and possibly better infrastructure but might increase travel time slightly.
- **Local Diversions:** Familiarity with local roads can help bypass congested or problem areas.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits Required:** Ensure all necessary permits are acquired for transporting hazardous materials.
- **Regulation Compliance:** Adhere to regulations regarding transport timings, speed limits, and placarding standards for hazardous materials.

7. Historical Incidents

There have been instances of road accidents involving heavy vehicles, often resulting from overloading, poor road conditions, or driver fatigue. There are strict checks to prevent accidents involving hazardous materials.

8. Environmental Considerations and Sensitive Areas

- **Natural Reserves:** Be cautious around any protected areas (forests, wildlife sanctuaries) as these may require compliance with specific environmental regulations.
- **Pollution Concerns:** Avoid spills or leaks, as the area may include agricultural lands sensitive to contamination.

9. Analysis of Communication Coverage

Communication networks generally cover most of the route; however, some rural areas may experience weak signals or dead zones, particularly in less populated stretches.

10. Estimated Emergency Response Times

- **Urban Areas:** Emergency response (ambulance, fire services) is relatively quick (20–30 minutes) within larger towns.
- **Rural Sections:** Response times lengthen considerably (30–60 minutes) in remote areas due to distance and accessibility.

11. Overall Summary of Risk Assessment

- **Risk Level:** Moderate due to potential severe weather impacts during the monsoon, typical traffic hazards, and rural sections with communication blackspots.
- **Recommendations:** Train drivers thoroughly on local conditions and emergency responses. Encourage contingency planning for weather-induced disruptions and ensure vehicle safety checks, given the nature of hazardous material transport.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	Medium	11.38342, 77.89975	30 KM/Hr
6	Turn	Medium	11.38190, 77.90254	30 KM/Hr
7	Turn	High	11.38186, 77.90258	15 KM/Hr
8	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
15	hospital	Aruthra Hospital	11.2264666, 78.1631524	30 km/h	Medium
16	hospital	Rudhra Hospital	11.226569, 78.163216	30 km/h	Medium
19	clinic	Sri Ram Nursing Home, Namakkal	11.2259986, 78.1631944	30 km/h	Medium
20	clinic	Thangavel Skin Care	11.2253765, 78.1644957	30 km/h	Medium
21	hospital	Rajendran Children's Hospital	11.2251887, 78.1649142	30 km/h	Medium
22	clinic	Lung Care Centre	11.2276789, 78.1638586	30 km/h	Medium
23	hospital	Vignesh Hospital	11.224818, 78.1621315	30 km/h	Medium
24	clinic	Vishnu Ayurveda Clinic	11.224262, 78.164173	30 km/h	Medium
25	clinic	Dhanvanthiri Physiotherapy	11.2245256, 78.1638528	30 km/h	Medium
26	hospital	Venkat Hospital	11.2253645, 78.1646159	30 km/h	Medium
27	hospital	Shree Meenakshi Dental Hospital	11.225269, 78.164275	30 km/h	Medium
28	hospital	Annai Eye Hospital	11.2241645, 78.1641366	30 km/h	Medium
29	clinic	Chandra Test Tube Baby Centre	11.2241806, 78.1645931	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
31	clinic	Kumaran Poly Clinic	11.2233138, 78.1639212	30 km/h	Medium
32	hospital	Ilamurugan Hospital	11.2248399, 78.166586	30 km/h	Medium
33	hospital	Aravinth Hospital	11.2233336, 78.1672805	30 km/h	Medium
34	hospital	Aburvaa Ayurveda Hospital	11.2228807, 78.1640453	30 km/h	Medium
35	hospital	Kathiravan Hospital	11.226651, 78.167351	30 km/h	Medium
36	hospital	Vanitha Child Care & Skin Hospital	11.2243378, 78.16811	30 km/h	Medium
37	hospital	Sri Dhanvanthries Hospital	11.2241181, 78.1678614	30 km/h	Medium
38	hospital	Sri Nidhi Hospital, Namakkal	11.2255711, 78.1687291	30 km/h	Medium
39	clinic	Athithya Nursing Home	11.2237726, 78.1686447	30 km/h	Medium
40	hospital	Sri Ramana Hospital	11.223503, 78.1685433	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium